

PROBLEMS ON COMPOSITE MATERIALS

1.- A composite material has an epoxy matrix reinforced with 68% in volume of unidirectional carbon fiber. The density of carbon fiber is 1.79 g/cm^3 and of the epoxy resin is 1.20 g/cm^3 .

- What's the percentage in weight of the carbon fibers and the epoxy resin?
- What's the average density of the composite?
- If the average density of the composite is 1.65 g/cm^3 , what's the percentage in volume of the carbon fibers inside the material?
- Which of the two materials has a higher Elastic modulus? Calculate its value.

Data: $E_{FC} = 400 \text{ GPa}$ and $E_{\text{epoxy}} = 3 \text{ GPa}$. Assume isostrain conditions.

(Solution: a) 76%; b) 1.6 g/cm^3 ; c) 76%; d) 304.7 GPa)

2.- A composite material is planned to be manufactured using an epoxy matrix reinforced with continuous, longitudinally aligned fibers. The composite must contain a maximum fiber volume fraction of 40%, and the minimum values for the elastic modulus and longitudinal tensile strength are set at $5.5 \cdot 10^4$ and $1.2 \cdot 10^3 \text{ MPa}$ respectively. The available candidate reinforcement materials include glass fibers, carbon fibers, and aramid fibers. Determine which reinforcement material is best suited to meet the specified requirements.

Material	σ (MPa)	E (Gpa)
Epoxy Resin	69	3.1
Fiberglass	2000	72
Carbon Fiber	3500	350
Aramid Fiber	3500	124

(Solution: carbon fiber reinforced composite)

3.- A composite material has a 69.5% in weight of unidirectional graphite-HM fibers in an epoxy matrix.

- Estimate the density and elastic modulus of the composite.
- If a 50% of the graphite fibers are replaced with Kevlar fibers, estimate the specific elastic modulus and the specific tensile strength in the longitudinal direction in this new hybrid composite.
- Calculate the minimum volumetric fraction of the graphite-HM fibers so that the initial composite has more strength than the hybrid composite of section b).
- Do the same calculations as in (c) considering Kevlar fibers.

Data:

Material	Density g/cm ³	Elastic modulus GPa	Tensile strength MPa
Epoxy resin	1.23	3.5	70
Graphite-HM fiber	1.86	390	2400
Kevlar-49 fiber	1.44	131	3700

(Solution: a) 1.61 g/cm³, 235.4 GPa; b) 106.5×10⁶ m²/s², 1.24×10⁶ m²/s² c) 0.77; d) 0.49)

4.- Calculate the elastic modulus of a unidirectional composite material with epoxy matrix and Kevlar 49 fibers, where 63% of its volume is fibers and is under isostrain conditions. If the tensile strength of Kevlar 49 fibers is 3700 MPa and the tensile strength of the epoxy resin is 70 MPa, calculate the tensile strength of the composite material. What fraction of load is carried by the Kevlar fibers?

Data: $E_{\text{kevlar}} = 131 \text{ GPa}$; $E_{\text{epoxy}} = 3.5 \text{ GPa}$.

(Solution: a) 83.82 GPa; b) 2.36 GPa; c) 0.984)

5.- For an E-glass fiber reinforced polyester composite with continuous aligned fibers, with 40% vol fraction fibers:

- Determine the Modulus of elasticity in the longitudinal direction
- If a cylindrical rod 19 mm in diameter of this composite is stressed in the longitudinal direction to 50 MPa, determine the forces in the fiber and matrix phases, respectively.
- Estimate the specific tensile strength of the composite.
- If the rod was originally 1.5 m long, estimate its extension under the applied stress.

Data: $E_m = 3400 \text{ MPa}$; $\sigma_m = 69 \text{ MPa}$; $\rho_m = 1.3 \text{ g/cm}^3$; $E_f = 72000 \text{ MPa}$, $\sigma_f = 3500 \text{ MPa}$; $\rho_f = 2.5 \text{ g/cm}^3$.

(Solution: a) 30.8 GPa; b) 13212N; 964N; c) 809.8 kN m/kg; d) 2.44 mm)

6.- Calculate the tensile modulus of elasticity of a unidirectional carbon-fiber-reinforced plastic composite material that contains 64 percent by volume of carbon fibers and is stressed under isostrain conditions. If the tensile strength of the carbon fibers of the 64 percent carbon-fiber-epoxy composite material is $0.31 \times 10^6 \text{ psi}$ and that of the epoxy resin $9.20 \times 10^3 \text{ psi}$, calculate the strength of the composite material in psi. What fraction of the load is carried by the carbon fibers? The carbon fibers have a tensile modulus of elasticity of $54.0 \times 10^6 \text{ psi}$ and the epoxy matrix a tensile modulus of elasticity of $0.530 \times 10^6 \text{ psi}$.

(Solution: a) 1.39 GPa; b) 0.995)

7.- A composite material is composed of an epoxy matrix and 68% in volume of unidirectional continuous carbon fibers. Calculate:

- The average density of the composite material.
- The weight percentage of the carbon fibers and the epoxy resin in the composite.
If the composite material is substituted by an Aluminium alloy AL2024, calculate:
- The young's modulus of the composite material and the percentage change of the young's modulus with respect to the aluminium alloy. Assume isostrain conditions
- The weight change for a cylindrical bar of material with 1 cm^2 cross-sectional area and 1 m in length.

Material	Density (g/cm ³)	E (GPa)
Epoxy resin	1.2	3.0
Carbon fiber	1.79	220
Al2024	2.28	73.4

(Solution: a) 1.601 g/cm³; b) 76% fiber-24% resin; c) 105.1%; d) 67.9 g lighter, -29.8%)

8.- Five kilograms of continuous boron fibers are introduced in a unidirectional orientation into 8 kg of an aluminum matrix. Calculate

- The density of the composite.
- The modulus of elasticity parallel to the fibers.
- The modulus of elasticity perpendicular to the fibers.

Data: $E_{\text{Al}} = 70 \text{ GPa}$; $\rho_{\text{Al}} = 2.7 \text{ g/cm}^3$; $E_{\text{Boron}} = 380 \text{ GPa}$; $\rho_{\text{Boron}} = 2.3 \text{ g/cm}^3$.

(Solution: a) 2.530 g/cm³; b) 201.1x10³ MPa; c) 106.4x10³ MPa)

9.- We want to produce 5 kg of a continuous unidirectional fiber-reinforced composite of HS (High Strength) carbon in a polyimide matrix that has a modulus of elasticity of at least $172 \times 10^3 \text{ MPa}$ parallel to the fibers. How many kilograms of fibers are required?

Data: $E_{\text{carbon fiber}} = 276 \times 10^3 \text{ MPa}$; $E_{\text{PI}} = 2070 \text{ MPa}$; $\rho_{\text{carbon fiber}} = 1.75 \text{ g/cm}^3$; $\rho_{\text{PI}} = 1.39 \text{ g/cm}^3$.

(Solution: 3.37 Kg)

10.- An epoxy matrix is reinforced with 40 vol% E-glass fibers to produce a 2-cm-diameter composite that is to withstand a load of 25000 N that is applied parallel to the fiber length. Calculate the stress acting on each fiber.

Data: $E_{\text{glass fiber}} = 72.4 \times 10^3 \text{ MPa}$; $E_{\text{matrix}} = 2.8 \times 10^3 \text{ MPa}$.

(Solution: 188 MPa)

11.- A titanium alloy with a modulus of elasticity of 110×10^3 MPa is used to make a 450 kg part for a manned space vehicle. Determine the weight of a part having the same modulus of elasticity parallel to the fibers, if the part is made of:

a) Aluminium reinforced with boron fibers.

b) Polyester (with a modulus of 4500 MPa) reinforced with high modulus carbon fibers.

c) Compare the specific modulus for all three materials.

Data: $\rho_{Ti} = 4.507 \text{ g/cm}^3$; $E_{Al} = 70 \text{ GPa}$; $\rho_{Al} = 2.699 \text{ g/cm}^3$; $E_{Boron} = 380 \text{ GPa}$; $\rho_{Boron} = 2.36 \text{ g/cm}^3$; $E_{carbon \text{ fiber}} = 530 \times 10^3 \text{ MPa}$; $\rho_{carbon \text{ fiber}} = 1.9 \text{ g/cm}^3$; $\rho_{polyester} = 1.28 \text{ g/cm}^3$.

(Solution: a) 265 kg b) 140.3 kg c) Ti: 24406 kN m/kg ; B-Al: 41431 kN m/kg ; C-PE: 78314 kN m/kg)

12.- A tungsten matrix with 20% porosity is infiltrated with silver. Assuming that the pores are interconnected, what is the density of the composite before and after infiltration with silver? The density of pure tungsten is 19.25 g/cm^3 and that of pure silver is 10.49 g/cm^3 .

(Solution 17.5g/cm³)